

AI-Powered Intrusion Detection for Wireless Networks

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Background

Wireless networks allow for enhanced capabilities and flexibility for individuals as devices can connect without any physical hardware. Mobility is therefore prioritized on wireless networks and is leveraged by professionals in the field of cybersecurity as well as threat actors looking for unauthorized access to data. Traditional methods of packet inspection, rogue access points, Man-In-The-Middle (MINM) and Denial-of-Service (DOS) attack vectors are ways that exploit network vulnerabilities which can be applied in the scenario of wireless networks. With the advancement of Artificial Intelligence (AI) technology and Machine Learning (ML) algorithms, threat actors can craft various network attacks exponentially easier with fewer resources than ever before. It is critical that network defense methodologies/tactics integrate AI as a way of combating wireless attacks.

An Intrusion Detection System (IDS) model can be used with AI technology to help secure a wireless network. It uses continuous monitoring of network traffic to report and find any malicious activity and suspicious behaviors that may be presented in a network. IDS employs signature-based, anomaly-based, and even hybrid-based techniques for detection purposes. These various approaches are covered in-depth within the results of an isolated wireless

network experiment that shows both the Supervised Learning (SL) and Reinforcement Learning (RL) methods of AI IDS modeling to secure a network against an attacker.

Introduction

Wireless networks allow for enhanced capabilities and accessibility for users to connect many personal devices. The delivery of data (IEEE 802.11) in such a network must be prioritized as transmissions through radio waves (2.4 GHz / 5 GHz) can be altered by threat actors. Using methods of AI and its applications of ML, IDS models can be developed to inspect packets.

This paper will examine the differences in security between the already established Wi-Fi security protocols to determine what attacks can be used in an isolated environment approach when discussing the methodology. The analysis will explore the various ML algorithms that are popular and give the best results concurrently. An experiment will then be covered using an isolated private wireless network and a list of tools that were used to produce results. Considerations and recommendations along with the topic of Deep Learning (DL) is discussed for future reference.

Objective

Develop an IDS model that can detect wireless traffic within a wireless network using methods of AI and ML. This is done with the following bullet points:

- Identify tools and techniques needed for wireless network testing.
- Describe the methodology of the project/experiment.
- Understand how to gather and preprocess data.
- Compare/Contrast SL and RL IDS models.
- Look at future tools and techniques for the AI IDS model approach.

Methodology

1. Create attacker data using Aircrack-ng
 - a. Within the isolated network, simulate attacks (such as deauthentication, fake authentication, 4-Way WPA2 Handshake capture)
 - b. This is possible with a Kali Linux Virtual Machine (VM) and a USB Wi-Fi adapter
2. Use Wireshark/Kismet for data collection using Packet Capture (PCAP) files
 - a. Wireshark is an open-source network packet analysis tool that allows for Wi-Fi data to be shown and saved in files called PCAPs.
 - b. Kismet is similar to Wireshark as it detects network traffic within a specific interface and is overall easier to find alerts when attacks are presented in a network
3. Extract important features from PCAP data
 - a. Features are the many pieces of data that can be used within a wireless packet. This is done by inspecting Wireshark data and converting values in a Comma-Separated-Values (CSV) file.

- b. Some features give more critical data than others (such as packet types, subtypes) to give evidence as to what is considered normal or malicious data.
- 4. Use AI IDS models to predict attacker data
 - a. This is done using Supervised Learning (SL) and Reinforcement Learning (RL) algorithms paired with Machine Learning (ML) and AI to predict malicious attacker data.
 - b. The SL and RL models are compared with visuals within the results of this paper.
- 5. Send data to Graylog for further visualization
 - a. In most Security Operation Center's (SOCs), IDS models are visualized with Security Information Event Management Software to easily filter out data and find the root cause of an attack.
 - b. Graylog is a powerful open-source Security Information and Event Management (SIEM) security solution that is used for log collection and analysis of data at the end of the experiment.

Overview of Machine Learning Algorithms and Datasets

There are many different Machine Learning (ML) models used throughout the field of Artificial Intelligence (AI) when making an Intrusion Detection System (IDS) model. The usage of ML has proven to be the most powerful security tool when detecting attacks and understanding the semantics behind systems and even zero-day attacks (anomaly detection). The main approaches to ML are Supervised Learning (SL) and Reinforcement Learning (RL). SL can be taught through a teacher-student relationship in which the training dataset trains the target dataset and classifies the labeled dataset [10]. Overall, SL is trained on known labeled data and learns to predict an output based on give inputs of the data. Large Language Models (LLMs) such as ChatGPT use versions of SL their data when predicting what the LLM outputs for a given user. RL on the other hand, aims to find an optimal learning policy strategy during the model-building phase [9]. This is done through an entity called an AI agent that makes decisions based on the data shown. An agent is usually learned by using optimal behaviors through a trial-and-error system and receives a reward as feedback from the agent. LLMs such as DeepSeek use RL in their ML algorithms to reduce resources needed for AI models and provide a more optimal approach when receiving output from the model.

Without getting into the mathematics of each ML algorithm, the most popular ones can be discussed at a high level with definitions. Supervised ML for intrusion detection has many common machine algorithms. The aim of SL is to achieve a goal based on the training samples given. Common tasks that are prevalent in this style of learning are classification and regression mechanisms. A Support Vector Machine (SVM) is a classification technique that yields state-of-the-art performance in classification and prediction and is one of the key models in ML research. The idea of SVM is to construct a high-dimensional feature space from input vectors, and sperate them into two classes, namely positive and negative. Positive data represents normal data whereas negative data represents abnormal data. Logistic regression is another supervised algorithm that is primarily used for both binary and multiclass classification tasks. The principle behind this algorithm is to predict the occurrence of an event based on the concept of probability [10].

K-Nearest Neighbor (KNN) is the simplest supervised ML algorithm to integrate in an IDS model as it can easily learn complicated approaches quickly. It classifies data based on the Euclidean distance measurement. The Bayesian classification algorithm calculates the probability to determine the likelihood using the Bayes Theorem. This is based on the idea that the probability of an event is believed to be true given some evidence. The random forest algorithm relies on multiple decision trees with each decision tree making predictions of the data independently. Random forest is more advantageous in terms of classification than other models when using a variety of different datasets when testing on a ML-based IDS model [10,11].

For the relevant datasets in the literature of IDS models it is skewed towards more towards physical networks rather than wireless networks due to its prevalence. The Knowledge Discovery and Data Mining Tools Competition (KDD-CUP) dataset is a predictive model capable of distinguishing between intrusions or attacks and normal connections using many input vectors. This dataset is useful for algorithms such as SVM, KNN and Bayesian classification. The NSL-KDD dataset improves the KDD-CUP data by removing duplicate records and increasing the proportion of samples in the testing data which can give better accuracy for most models when integrating intrusion detection. This dataset is used mainly in logistic regression and random forest to produce the best predictions of data [10]. The Canadian Institute for Cybersecurity Intrusion Detection System (CICIDS) dataset contains the most up-to-date common attacks with network captures of traffic and system logs of many different operating systems. A study was shown using the random forest algorithm alongside the CICIDS 2018 dataset where the IDS showed 98.5% accuracy and 100% accuracy respectively [11]. For wireless networks, the Aegean Wi-Fi Intrusion Dataset (AWID) is the most used for wireless IDS data. It is the latest and most effective publicly available dataset for wireless network intrusions with many attributes and wireless attacker data. The problem with AWID is its ineffective data imbalance which requires a significant amount of data preprocessing methods to be used for research [4].

With the supervised ML algorithms explained and relevant datasets discussed, there are many other reinforcement ML algorithms that are prevalent in the research. For the purposes of this paper, the relevant algorithms to be discussed and tested within the experiment will be Deep Q-Network (DQN), Proximal Policy Optimization (PPO) and Group Relative Policy Optimization (GPRO). DQN integrates deep neural networks into the decision-making process which is part of the concept of Deep Learning (DL). This makes the algorithm possible to solve complex tasks using what is called Q-learning. This is a model-free method that learns an optimal policy by estimating the Q-value function which gives back a reward for the AI agent after a specific action is processed in each state. It is shown that DQL along with a Multi-Agent Feature Selection (MAFS) method to improve specific data attributes and vectors for network attack classification results in high accuracy rates when using datasets such as CICIDS and NSL-KDD [8]. For the PPO algorithm, the AI agent is adjusted to maximize expected rewards and constrain the policy for each state when the agent receives a reward to avoid any large changes to the model. This is further expanded upon with DeepSeek's GRPO algorithm to save training data costs of RL. GPRO examples a group of outputs from the old proximal policy and then optimizes the policy model by maximizing hyper-parameters that are computed to produce

outputs within the model's architecture. This of course comes with a reward modeling system as it decides the optimization direction of RL [6].

Wireless Protocols and Overview of Attacks/Defenses

Before the development and tools needed for the experiment, definitions of the different wireless protocols must be established to produce attacker data. Every wireless network is enforced with a security mechanism and there are different levels of this mechanism. Wired Equivalent Privacy (WEP) was the sole security mechanism in the first version of the 802.11 protocol introduced in 1999. The main goal of WEP was to provide confidentiality level comparable to wired networks with authorization and traffic encryption. The authorization aspect of WEP supports two methods of authentication called open system and shared key. For open system authentication the client does not need to provide any credentials for connection to the Access Point (AP). The authentication is completed after the exchange of only two messages. In the shared key authentication, a process that completes with the exchange of four messages takes place before a client can enter the network: (1) the client sends an Authentication Request message which contains the Media Access Control (MAC) address of the client and the MAC address of the AP, (2) the AP responds with a challenge message which contains a 128 bits random number, (3) the client sends a response message which contains the random number encrypted with the WEP shared key. The AP then decrypts the previous message using its shared key. If the number contained in the decrypted message matches the random number previously sent, then the AP considers that the client is in possession of the shared key. The final message (4) is an Authentication Response message containing the outcome of the authentication process. The message exchange of WEP proves that it is strictly unidirectional, meaning the AP can authenticate the client but not vice versa and is critical for understanding WiFi Protected Access (WPA) and WPA2 protocols [5].

The traffic encryption application of WEP depends on the Rivest Cipher 4 (RC4) algorithm which is a symmetric-key stream cipher algorithm and is used for the generation of a session key. This type of encryption will not be covered in-depth as it is not used for direct packet encryption which is used in WPA and WPA2 networks. WPA was introduced to alleviate the weaknesses WEP presented. Stronger encryption methods were introduced such as the Advanced Encryption Standard (AES) and addressed critical security issues of the Extensive Authentication Protocol (EAP) in packets and increased Initialization Vector (IV) lengths for stronger integrity checks. WPA depends on central authentication servers such as Remote Authentication Dial-In User Service (RADIUS) for user authentication, access control and management. A variation of WPA was also created called WPA Pre-Shared Key (WPA-PSK) for a more simplified version that uses a passphrase as a pre-shared secret key among the users such as in the case of WEP [5].

WPA2 (IEEE 802.11i) is the most widely used protocol in many wireless networks today. It uses a single key which is placed at the highest level of the hierarchy when it comes to encryption. If the authentication method is based on a pre-shared key the key is simply referred to as the Pre Shared Key (PSK). If the version of WPA2 is based on the 802.1X framework (often used with 802.11i for authentication) the top-level key is called the Master Session Key (MSK). These keys are used for generating the primary keys in WPA2, which is the Pairwise

Master Key (PMK). There is even another level of the encryption key hierarchy as it is critical for confidentiality and integrity. These keys are referred to as the Pairwise Transient Key (PTK) and the Group Transient Key (GTK). They are specific to the client-AP pair as they are produced during the authentication process from the PMK. WPA2 uses three alternative protocols for protecting network traffic. Temporal Key Integrity Protocol (TKIP), Counter-Mode/Cipher Block Chaining Message Authentication Code Protocol (CCMP) and wireless Robust Authenticated Protocol (WRAP). TKIP is based on RC4 which is a transitional step from WEP. WRAP, on the other hand, is based on the Offset Codebook (OCB) mode of AES which is more secure but can be subject to licensing issues of the network. CCMP is based on the AES algorithm in the Counter with Cipher Block Chaining-Message Authentication Code (CCM) mode. It breaks the plaintext in chunks of 128 bits and encrypts them with a key of the same size [5]. It's also important to mention WPA3 is the latest Wi-Fi security standard as it offers enhanced encryption with a 192-bit encryption rather than WPA2 128-bits encryption. Even though it is generally considered more secure than WPA2, the WPA2 protocol remains widely used as it is still extremely difficult to obtain root keys with strong passwords and methods of packet/data encryption.

There are many different attacks throughout wireless network IDS research. For the purposes of the experiment described below, the focus will be on WPA/WPA2 specific attacks as the wireless network tested will be using WPA2. A dictionary attack is a form of brute force that has been widely used to retrieve weak WPA/WPA2 keys. This is considered the most reliable method for WPA/WPA2 cracking. The first phase of the attack is the attacker sniffing a network to catch a live handshake between a client device and AP. In the second phase, the attacker goes through a process of generating the third message of a 4-way-handshake based on potential keys in a large database or dictionary. For each key in the dictionary, the attacker evaluates the result against the captured sequence and if the two keys match, the attacker can be certain that the PSK can be obtained on the wireless network being attacked. For this attack to prove successful, the dictionary must have the passphrase of the network stored already in its database [5]. This can be relatively difficult to achieve if the wireless network has strong passwords or Multi-Factor Authentication (MFA) enabled by default.

Availability attacks commonly referred to as Denial of Service (DoS) attacks, are prevalent in wireless networks as well. A deauthentication attack is the easiest and most efficient for an attacker to achieve. It is due to the fact that deauthentication packets are transmitted unprotected and can easily be spoofed by fake packets within the network. Upon receiving such a packet, the client must abandon the network immediately without any additional actions required. The attacker monitors the traffic on the network to find the MAC address associated with a specific client and the one of the AP. A deauthentication management frame can then be formed and sent to that client on behalf of the AP. This can be also sent to the AP on behalf of the client, so that the network stops considering that client authenticated. A disassociation attack is similar to deauthentication. A disassociation message is sent between a client and AP as the spoofed packet leading to a duration of a loss of service. To protect from such attacks, using mac address filtering and AP isolation can hide client MAC address and restrict access to a wireless network [5].

Although the experiment will not cover Man-in-the-Middle (MITM) attacks, it is critical to understand as IDS models can be used to pick up on these attacks using AI and ML. An Evil

Twin is a special case of a Honeypot (networks created and controlled by malicious administrators) that advertised an existing Extended Service Set Identifier (ESSID) or Service Set Identifier (SSID) to trick users into connecting to it instead of the valid network. Evil Twin APs are possible because multiple APs with the same ESSID are allowed to exist in the same area and in some cases, the client will prefer to connect to the one with the strongest signal disregarding the BSSID of the legitimate AP. This is common to impersonate networks that are open, such as coffee shops and airports. There are also Rogue APs which are unauthorized access points (hardware or software) enabled within the corporate, home or office premises by an insider of that network. They are usually connected to the wired counterpart of the network, but wireless connection is not uncommon especially for software APs [5]. To defend against these attacks, disabling ESSID or SSID broadcasting along with MAC filtering and a strong signal can reduce visibility to Evil Twin attacks. Rogue APs are more challenging to detect and mainly have to do with how a network is designed. Securing physical access and using Virtual Private Networks (VPNs) among other detection tools can prevent rogue APs.

Results

Create a Wireless Network Within an Isolated Environment

Now that the wireless protocols and main attacks/defenses are presented, an experiment can be conducted within an isolated environment. For an outline of the approach to this research experiment, a brief list of prerequisites and tools is therefore provided:

Prerequisites for the experiment:

- **USB WiFi Adapter – ALFA AWUS036ACS:** This is used to send attacker data to the AP (router) from a Kali Linux Virtual Machine (VM).
- **Router – NetGear AX1800:** The AP that will be used as an isolated/private wireless network.
- **Laptop:** A laptop is required for the connection of a USB WiFi Adapter, any VMs needed and the creation of AI IDS models.

Tools used for the experiment:

- **Python:** Used for development of IDS models using libraries for ML integration with SL and RL methods.
- **Jupyter Notebook:** Project Jupyter is a project to develop open-source software, open standards, and services for interactive computing across multiple programming languages. This is used with a custom Python kernel for wireless network sniffing and data collection.
- **Wireshark:** Used to extract data into PCAP files and produce normal and malicious data.
- **Kismet:** Similar to Wireshark, Kismet allows for alerts to be seen easily when malicious data enters a network. Kismet will verify any attacker data is successfully presented in the experiment.
- **Graylog:** An open-source SIEM that allows for data to be visualized on a broader scale.

- **Aircrack-ng:** A complete suite of tools to assess WiFi network security. The specific tools within Aircrack-ng are used below:
 - **airodump-ng:** Used for packet capture, capturing raw 802.11 frames.
 - **airmon-ng:** Used to enable monitor mode on wireless interfaces (ALFA AWUS036ACS).
 - **airreplay-ng:** Used to inject frames. This is used to conduct deauthentication, dissociation and any other relevant attacks in the experiment.

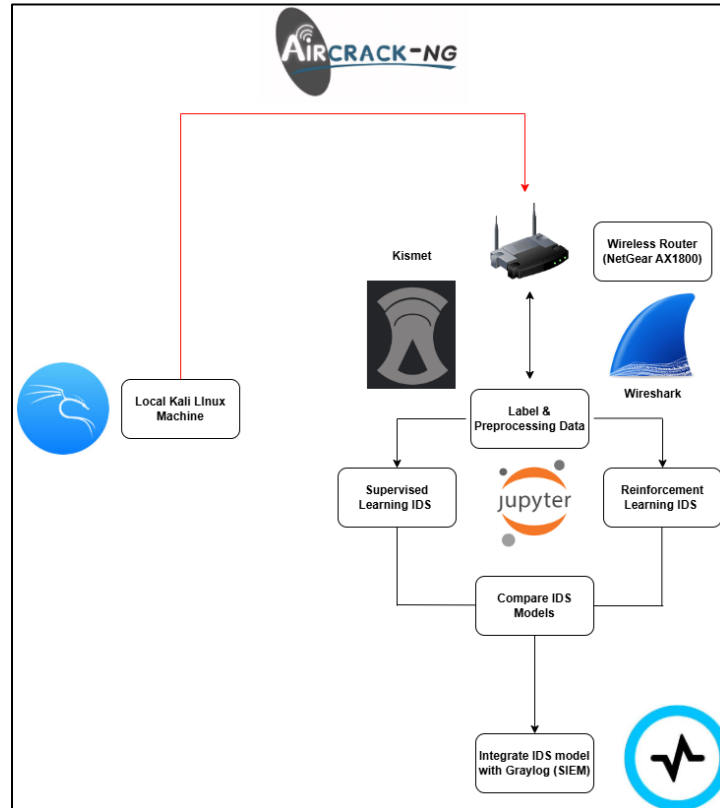
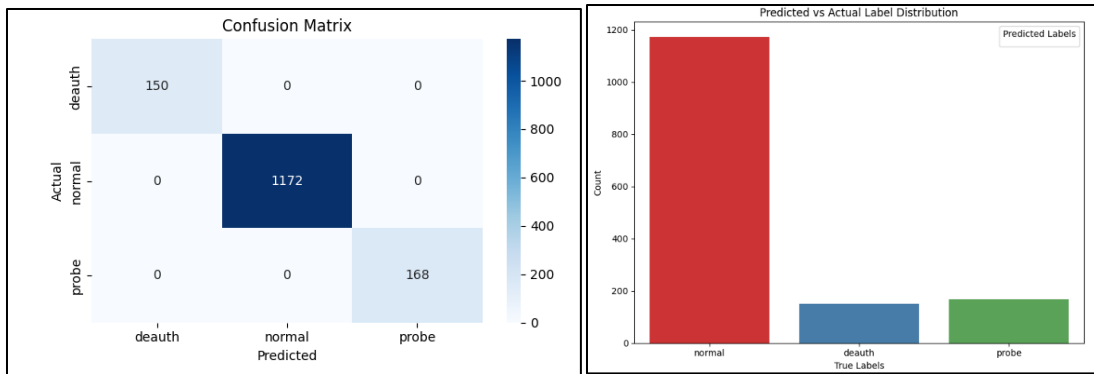


Figure 1: High-level Overview of AI IDS environment

The methodology of this paper goes over the steps outlined in this section as well as a high-level overview of the environment's prerequisites and tools. Another important note is that the wireless network will use the WPA2 protocol with AP isolation turned off to find clients easily on the network. The only client connected to the private wireless network is a personal iPhone 14 Pro to establish a 4-way handshake between the client and the AP and vice versa. Attacks used within the network are the DoS common attacks established above using Aircrack-ng packages [3].

Comparative Analysis of Supervised and Reinforcement Learning with Recommendations

For the findings of this research, Random Forest and SVM are used for the SL model and DQN for RL model. The current IDS models do not use any current datasets such as AWID for the results and only using the input data given from PCAP files. For the SL model, random forest provided 100% accuracy and 85% accuracy for the SVM.



Figures 2 and 3: Random Forest IDS results

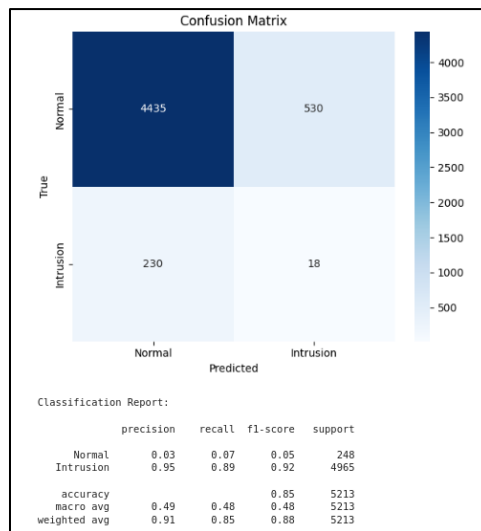
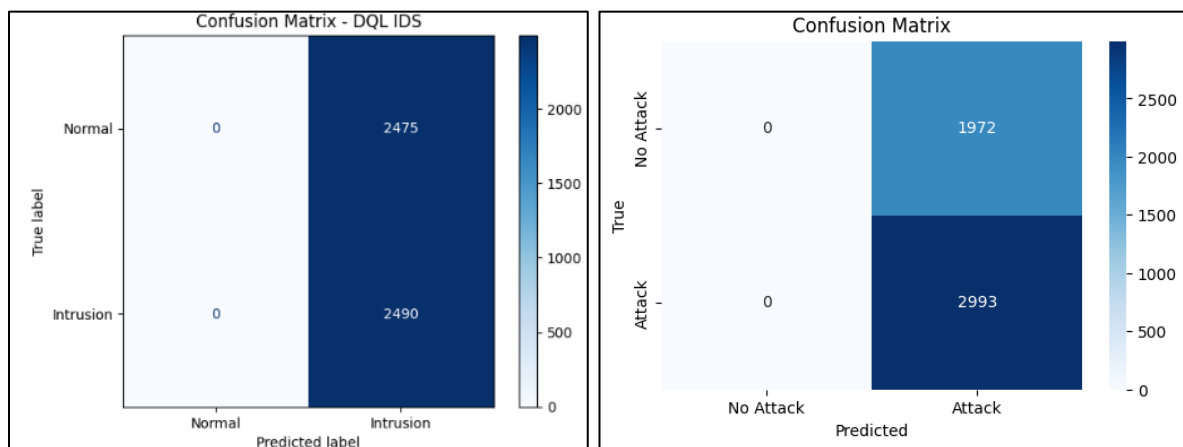


Figure 4: SVM IDS results

Currently the model uses only the input data that is given and not predefined training datasets for enhanced learning. This works with the SL model but not with RL data.



Figures 5 and 6: DQN IDS results

The above shows incorrect outputs of true normal and attacker data for the RL model. The problem with RL is the difficulty of developing an AI agent with an optimal reward system to train the data. Recommendations to be considered are to use online resources such as [Orange](#) for SL as it is easy to visualize data using drag and drop methods with many ML models. For the RL side it's more complex as there are not many open-source resources for AI agent deployment. Future research and development must be conducted to provide adequate results for RL modeling.

Deep Learning Integration and Next Steps

As IDS research becomes more effective, Deep Learning (DL) offers greater accuracy than conventional ML methods. DL can handle raw facts and pick up high-level characteristics independently. Although classic ML techniques are successful in detecting intrusions, they are constrained by the necessity to create sample characteristics artificially. Deep Neural Networks (DNN), which include numerous hidden layers with nodes and ways to link them, are the one of the most significant computational networks to be used in IDS models to achieve great results [2]. DL can be applied to both SL and RL methods. From a traditional supervised ML approach [1] to applying DNN to SL, more complex features can be selected and trained on feature-rich data. A research study has also shown that Deep Reinforcement Learning (DRL) can solve many complex practical problems using neural networks in IDS models [7]. Although DL offers a more powerful approach to intrusion detection, it can be difficult to implement entirely without the required resources. Further research must be applied to IDS models to enhance techniques of feature selection, data preparation and preprocessing.

For the next steps of this research and experiment data, developing methods of deep learning apply greatly to produce results for supervised and reinforcement ML models. Other future steps that need to be integrated is live packet traffic. Currently, the methodology is to gather the data and save the data to a PCAP file in Wireshark. Automating or scripting attack vectors will allow for faster exposure of malicious data. Expanding the features of the data will produce greater overall results due to many of the packet headers in wireless traffic. Using Graphics Processing Unit (GPU) passthrough rather than a Central Processing Unit (CPU) for large amounts of data will make IDS models run smoother and faster. This will prove significant when applying datasets such as AWID into the experiment. Also, running the wireless network on the local machine as opposed to a VM will make gathering data exponentially faster. Currently, the USB WiFi adapter works on Kali Linux only with later versions of Windows as well. To make this work, more space on the local machine is required for a Dual Boot of a Linux operating system.

Conclusion

In this paper, the findings of this research provide evidence of successful network traffic with a WPA2 encrypted network. Using AI IDS models gives results with SL more than RL due to the difficulties of producing a working AI agent. Using SVMs on known data and other alternatives can give solutions to predict anomalies with RL IDS models in the future based on the experiment. Compared to traditional methods of ML techniques, using DL yields greater results with DNNs and significant feature extraction techniques with supervised and

reinforcement ML models. To meet the growing need for security in intrusion detection, AI-based IDS models for malicious and unknown attacks are formed based on the currently known attacker data and defenses. Datasets are formed to compare data using real-world scenarios based on what has been confirmed as valid attacker data in the past. For the future, AI technologies and its applications must be used in environments where sensitive data in present and research in isolated environments using private networks for enhancement of data performance and accuracy.

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